

Carina (Yongyan) Zheng

SENIOR AI ENGINEER, DATA SCIENTIST | PROJECT LEAD | AGILE LEAD | SCRUM MASTER

Auckland, New Zealand

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Key Skills

AI ENGINEERING

- Build commercial AI systems for embedded devices and the cloud, by using **Python** with different **frameworks (Torch, TorchVision, TensorFlow)** and libraries (**OpenCV, Numpy**)
- Develop and maintain customized and pre-trained **object detection** models (YOLO, SSD, Faster-RCNN)
- Implement efficient solutions with CPU, **NVIDIA products (DeepStream, CUDA, TensorRT, Jetson GPU)**
- Implement **AI optimizations** throughout developments
- Develop **data pipelines** and ETL workflows that perform tasks such as data engineering, wrangling and processing
- Evaluate solutions with performance metrics and cost-benefit measurements
- Provide insightful and professional advice based on comprehensive research
- Suggest and develop **prototype products** for senior managers and stakeholders to generate business values
- Develop **AI Android apps** and monitor all costs

DEVELOPMENT

- Develop in **Linux** environment, capable with **x86-64, ARM, AARCH** architectures and **Docker**
- Professional and clean-code OOP development experience
- Strong quality assurance awareness. Excellent capability to complete development by implementing detail-orientated pipeline design, followed by automated testing with **unit test, Jenkins, CT/CICD**
- Version control by using **Git (GitHub, BitBucket)** and Data Version Control (**DVC**)

SOFT SKILLS

- Business Savvy, expert in communicating with clients to tailor the solutions to meet their business needs
- Ability to interact professionally with technical and non-technical team members
- Good verbal, written communications and interpersonal skills, relating to a range of different personality types
- Proficient with JIRA, SCRUM and AGILE. Able to manage time effectively to ensure tasks are completed in time

Employment History

Int. Data Scientist

Auckland, New Zealand

PLEXURE

Oct 2023 - Current

Build ETL data pipelines among various platforms

road

Auckland, New Zealand

EROAD

July 2022 - Oct 2023

- **AI Engineering:** Computer Vision Achievements
 - Develop a cloud-based DMS, features include detecting mobile usage, blurring drivers' faces
 - Develop a real-time Driver Monitoring System (DMS) for Android devices
 - Develop entire module data pipeline to process RGB, thermal and video data
 - Develop corresponding E2E testing and production AWS pipelines
 - Establish and lead the coding standard implementation within the department
 - Work closely with stakeholders to deliver company's and clients' business values
 - Establish an AI communication forum within EROAD
- **Senior Data Scientist:** Modelling, Prediction and Data Visualization
 - Develop prediction model to predict
 - Build Data Visualization reports by using PowerBI, Shiny

Machine Learning Engineer

Auckland, New Zealand

CUBIC DEFENCE NEW ZEALAND

Nov 2020 - July 2022

- Develop a real-time object detection application on Jetson NX in a project team environment
- Design and implement project data capturing task on the field by using RGB and thermal cameras
- Plan, research and execute cutting-edge domains' solutions (e.g. in algorithms, optimization, GPU utilization, pipeline) to advance client's needs
- Use Python, Torch and NVIDIA developer products to train, evaluate, deploy and maintain customized models
- Build and deploy corresponding Docker images to meet development requirements
- Prototype hardware and software solutions for modeling and detection
- Research and prototype techniques and algorithms for object detection on Jetson and Diamond systems
- Contribute to preparing various product demonstrations for clients

Data Scientist

Christchurch, New Zealand

INTELLIGENZ SOLUTIONS

Nov 2019 - Feb 2020

- **Data Visualization:** PowerBI reports and dashboard
 - Understand clients' businesses and design key business questions to be solved
 - Design reports by users' roles and needs to show related information
 - Build interactive reports to assist with making informed business decisions
 - Build real-time dashboards for the management team to make faster decisions
- **Machine Learning: Automated prototype system**
 - Perform Exploratory Data Analysis to understand client's data
 - Design the system architecture
 - Build a **classification model** to predict users' behaviours
 - Reports prediction result to clients via PowerBI reports
 - Report project progress to manager and client regularly
 - Coordinate with technical lead to finish project environment setup

Data System Analyst

Christchurch, New Zealand

STATISTICS NEW ZEALAND, WWW.STATS.GOV.TZ

Oct 2016 - Oct 2020

- Data Custodian (secondment to Data Ventures, dataventures.nz)
 - Develop and manage data pipelines for large dataset
 - Efficiently discover and derive hidden information from the datasets
 - Perform data wrangling tasks on existing and new data
 - Cooperating with commercial banks to provide financial provision to banks, government departments and the general public by using R and AWS
 - Work in a fast-paced and agile environment that is responsive to customer needs
- Project-base and Organization-wide duties
 - Lead a small team to deliver project designs, implementations and technical development
 - Use Pentaho to build highly automated ETL workflows to process raw input data which reduces 90% of error
 - Design post-process storage and report system to provide clear views to front-end users
 - Co-work with external customers to enable go-live implementation
 - Design, build and maintain customized informative reports in Shiny, SSRS, SQL and PowerBI for projects
 - Provide platform training across the organization
 - Research, adapt and deliver new technologies to different platforms

Database Analyst and Support Technician

Christchurch, New Zealand

INTELLIGENZ SOLUTIONS, WWW.INTELLIGENZ.GLOBAL

Nov 2011 - Oct 2016

- Develop and enhance modules in the software in C#
- Debug code errors and implement hotfixes
- Train and support domestic and international clients on-site
- Manage clients' and company's databases and ensure they have high data integrity
- Integrate data from various database types into the software's required database structure
- Deliver business and technical support to customers
- In charge of software internationalization and localization, and applying quality control to multilingual resources

Education

Master of Applied Data Science (MADS), Distinction (GPA 7.3 out of 9)

Christchurch, New Zealand

UNIVERSITY OF CANTERBURY

2019 - 2020

- UC Representative, Airport Data Hackathon, hosted by Christchurch International Airport.
- Tutor of Statistics 101 and Statistics 401, MADS Social Club President.

Bachelor of Science, majors: Statistics and Computer Science

UNIVERSITY OF CANTERBURY

Christchurch, New Zealand

2014 - 2016

Summer Scholarship Project 2015/16 with Statistics New Zealand: Built a statistically informed interactive application to help residents understand how people commute to work.**Diploma in Information and Communications Technology**

CHRISTCHURCH POLYTECHNIC INSTITUTE OF TECHNOLOGY

Christchurch, New Zealand

2009 - 2011

Certificates

AGILE ICAgile Certified Professional (ICAgile), Business Value Analysis (ICAgile)**Other areas** First Aids (St John), Kaggle competition participant, Fire warden

Referees

Provided on request.